

# Murali Karthik Ganji

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Computer Science student at the University of Arizona pursuing minors in Business Administration and Statistics & Data Science, with experience in applied AI research, machine learning, data engineering, and software development. Experienced in developing Python-based data pipelines, research automation workflows, computer vision datasets, and machine learning applications. Strong interest in building practical AI-driven systems at the intersection of technology, data, and real-world decision making.

## EDUCATION

### University of Arizona, College of Science

*Bachelor of Science in Computer Science*

**Tucson, Arizona**

Expected Graduation Date: December 2026

- **Minor:** Business Administration, Statistics & Data Science
- **GPA:** 3.10
- **Honors:** Global Wildcat Scholarship, Dean's List Award – Spring 2024

**Relevant Coursework:** Data Engineering, Systems Programming, Software Development, Computer Organization, Machine Learning, Statistical Computing, Web Programming, Analysis of Discrete Structures, Natural Language Processing, Comparative Programming Languages, Linear Algebra, Vector Calculus

## TECHNICAL SKILLS

**Programming Languages:** Python, Java, SQL, JavaScript, C, R, Swift, Haskell, Prolog, Ruby

**Frameworks & Libraries:** React, Node.js, Express.js, Pandas, NumPy, scikit-learn, LightGBM

**AI, Computer Vision & Data Engineering:** OpenCV, FFmpeg, Supervisely, Image Processing, Semantic Segmentation, Data Pipelines, ETL, Feature Engineering, Data Cleaning, Statistical Modeling, Time Series Forecasting

**Databases:** MongoDB, MySQL, PostgreSQL

**Cloud & Developer Tools:** AWS, Docker, Git, GitHub, Bash, Linux/UNIX, VS Code

**Software Engineering:** Data Structures, Algorithms, OOP, Design Patterns, RESTful APIs, MVC Architecture, System Design, SDLC, Agile, Version Control, Testing, Debugging

**Office & Productivity:** Microsoft Excel, Word, PowerPoint, Outlook, Google Workspace

**Operating Systems:** Linux/UNIX, Windows, macOS

## PROFESSIONAL & RESEARCH EXPERIENCE

### Undergraduate Research Assistant (VIP Program – AI-Driven Healthcare Applications)

**Tucson, AZ**

*University of Arizona – Department of Electrical & Computer Engineering*

July 2026 – Present

- Develop Python-based preprocessing pipelines for retinal surgical video analysis and computer vision dataset preparation.
- Implement blur detection, SSIM-based filtering, metadata generation, and automated quality-control workflows to improve research dataset quality.
- Prepare annotated semantic-segmentation datasets and case-level data splits to support machine learning model development and evaluation.
- Collaborate with researchers using Git/GitHub, documentation, and reproducible development workflows.

### Web Development Intern

**Hyderabad, India**

*Susampada Developers*

May 2023 – July 2023

- Maintained and updated website content while improving usability and responsive performance.
- Assisted with SEO optimization, digital content management, and website maintenance.
- Coordinated with clients to gather project requirements, maintain documentation, and support web development initiatives.

## PROJECTS

### Forecasting Web Traffic for 145,000+ Wikipedia Pages (CSC 480 – Machine Learning Final Project) | [GitHub](#)

- Engineered scalable Python data pipelines to process and analyze traffic data for over **145,000 Wikipedia pages**, supporting feature engineering, model training, and evaluation.
- Designed lag-based and rolling-window features to capture temporal patterns and improve forecasting performance on sequential datasets.
- Implemented a hybrid **LightGBM-LSTM** forecasting model that improved **SMAPE accuracy by approximately 7%**.
- Conducted large-scale preprocessing, model validation, and reproducible experimentation for time-series forecasting research.

### Student Performance Prediction System (AI / Machine Learning Project)

- Built machine learning models to predict student academic performance using behavioral and academic datasets.
- Implemented **Decision Tree** and **Random Forest** algorithms for classification and regression tasks.
- Evaluated model performance using **MAE** and **R<sup>2</sup>** metrics to assess prediction accuracy and generalization.
- Analyzed feature importance to identify variables most strongly associated with academic outcomes.

### Research Data Engineering Project | [GitHub](#)

- Designed an end-to-end data pipeline to collect, clean, transform, and analyze datasets related to social media usage and student well-being.
- Automated ETL workflows using Python to improve data preparation and analytical efficiency.
- Performed exploratory data analysis and generated insights through data transformation and visualization techniques.

**CineTicket – Movie Ticket Reservation System** | [GitHub](#)

- Built a full-stack movie ticket reservation platform using **Node.js**, **Express.js**, **MongoDB**, and **RESTful APIs**.
- Implemented secure user authentication, session management, booking history, and database integration using **Express-session** and **Mongoose**.
- Developed a dynamic frontend using the **Fetch API** to support asynchronous movie browsing and ticket reservation workflows.
- Utilized Git/GitHub for version control, collaborative development, and project management.

**fintrackr – Financial Analytics & Statistical Computing Toolkit (R)** | [GitHub](#)

- Built an R-based financial analytics toolkit supporting budgeting, investment analysis, and debt evaluation.
- Implemented reusable analytical functions using **tidyverse** and **rlang** for financial summarization and comparative analysis.
- Created interactive visualizations with **ggplot2** to identify financial trends and support exploratory analysis.
- Applied statistical computing principles to develop maintainable and reproducible analytical workflows.

**Restaurant Management System – Collaborative Software Engineering Project (CSC 335)** | [GitHub](#)

- Collaborated in a multi-developer GitHub repository using **Agile**, pull requests, code reviews, version control, and testing workflows.
- Designed application components following the **Model–View–Controller (MVC)** architecture to improve maintainability and separation of concerns.
- Managed feature integration, branch management, debugging, and collaborative software development using Git.

**FinTrack iOS Application – Personal Finance Management App (Swift)** | [GitHub](#)

- Developed a personal finance management application using **Swift** and **SwiftUI** for expense tracking, budgeting, and financial reporting.
- Designed responsive user interfaces and implemented local data storage for transaction and category management.
- Applied object-oriented design principles to create maintainable and scalable application components.
- Integrated charts and interactive visualizations to improve usability and user experience.

**CERTIFICATIONS**

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- Machine Learning Specialization – Stanford University & DeepLearning.AI (2026)
- Advanced Learning Algorithms – Stanford University & DeepLearning.AI (2026)
- Supervised Machine Learning: Regression & Classification – Stanford University & DeepLearning.AI (2026)
- AWS Educate – Introduction to Generative AI Badge (2026)
- Artificial Intelligence Fundamentals – Great Learning (2025)
- Front-End Development (HTML) – Great Learning (2024)
- Infosys Catch Them Young (2021)

**LANGUAGES, SKILLS, ACTIVITIES & INTERESTS**

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**Languages:** English, Telugu, Hindi and French

**Soft Skills:** Leadership, Organization, Attention to Detail, Communication, Team Collaboration, Time Management, Problem Solving

**Activities:** Student Council, TECHNIX-2019, Youth Parliament-2019, Infosys Catch Them Young-2021

**Interests:** Software Engineering, Distributed Systems, Cloud Computing, Machine Learning, Open-Source Software, Scientific Computing